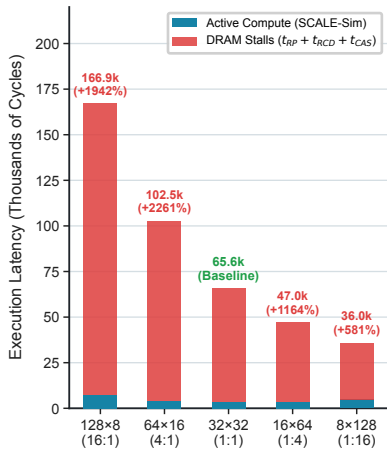
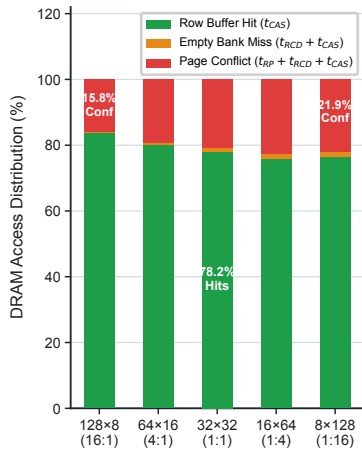


Closed-Loop Design Automation: Breaking the DRAM Command Timing Wall ($t_{RP} + t_{RCD} + t_{CAS}$)

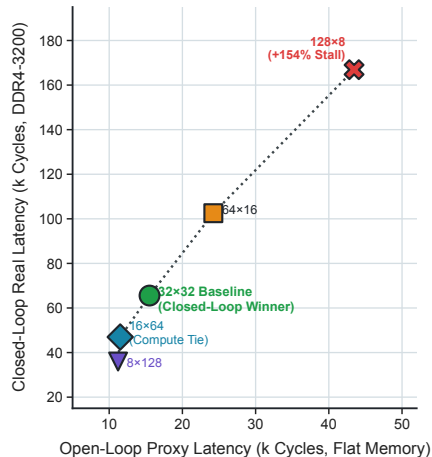
Panel A: Closed-Loop Latency Blowup



Panel B: Row-Buffer Conflict Breakdown



Panel C: Closed-Loop Pareto Inversion



PHYSICAL TAKEAWAY: Open-loop compute proxies hide the DRAM command timing wall ($t_{RP} + t_{RCD} + t_{CAS}$). While 16x64 and 32x32 tie in compute cycles (3,366 cycles), cycle-accurate DRAM simulation exposes severe memory stalls (+1,164% to +2,261%) and high conflict rates in asymmetric geometries, proving balanced aspect ratios dominate under real memory interfaces.